INTERVIEWEE: **Jenny Pickworth Glusker**

INTERVIEWERS: **Jacqueline Boytim**
Sarah Schneider

LOCATION: **Interviewee's home**
Huntingdon Valley, Pennsylvania

DATE: **1 November 2022**

[00:00:00]
SCHNEIDER: So today is Tuesday, November 1, 2022. My name is Sarah Schneider, and I am joined by Jacqueline Boytim. We are conducting the first session of an oral history interview with Dr. Jenny Pickworth Glusker at her home in Huntingdon Valley, Pennsylvania. So thank you for being with us today and participating in the oral history. And so to start off—

[00:00:21]
GLUSKER: Should I start out again? A little bit, yes.

[00:00:23]
SCHNEIDER: Yes, if you could start talking about some of your memories from growing up and tell us where you were born and where you grew up.

[00:00:30]
GLUSKER: Well, I was brought up in England, in Birmingham, which is a big industrial city. No real rivers, but a very good canal system, which is, you know, if you're a child, is of great interest to watch. And my father [Frederick Alfred Pickworth] and my mother [Jane Wylie Stocks] were both physicians at a mental hospital on the edge . . . almost in the country, just suburban Birmingham. Birmingham was built so they had circles of grass areas, and then more housing and then more circles of grass area. So it was rather well designed. And the buildings that he worked in were really nice, and they'd have beautiful gardens so the patients could walk there but couldn't get out. Then they met.

[00:01:31]
And my father's father [Alfred Joseph Pickworth] was a Wesley Methodist minister and a general practitioner, and he had . . . what did he have? He had six children [Florence "Flossie," William, Frederick Alfred, Thomas Claude, Eric, and Ralph], one of whom [Ralph] died at birth. The first three were by his first wife [Sarah Hannah Hart], who was really quite an artist. Somehow, several of us have just a little bit of ability to do artwork, which is nice when you

know that it's come from a particular relative. I'm not quite sure what else I can say. Anyway, here's a picture of my father. I don't know if you can turn that around.

[00:02:20]
BOYTIM: And you were telling us, his mother [Sarah Hannah Hart] died while he was quite young?

[00:02:24]
GLUSKER: Yeah, he was one year old, exactly, yes. His father didn't know what to do. So he called his sisters [Charlotte, Caroline, Jessie, and Emily] to come and help him look after the three children. They squabbled like mad. He sent a message—I have this from my cousin—he sent a message down to the village saying he was interviewing for a new wife and he had three children. Could they send people up? So they sent people up.

[00:02:54]
And what he did was, he gave them a cup of tea, very hot tea, in a saucer, and then he waited to see how they managed to arrange to drink it. It was too hot to drink, but did they put a little in the saucer or what did they do? And I'm not sure what this one wife [Mary Household] did, but she did okay, and so he married her and he had three boys [Thomas Claude, Eric, and Ralph] by her. And they lived in Lakenheath, [England], which no one had ever heard of until World War II, when it was a US Air Force base, Lakenheath. When his father had . . . his father had died, a lot of people would come to his house and knock on the door and there would be an American living in the house. And they'd say, "Is this Pickworth's old place?" Now if you go into town, into the village, you will see the big sign: "Pickworth's old place." So he was remembered as someone . . . .

[00:04:05]
So I had three uncles and one aunt from my father's side. But my father was very good at designing things, and the thing that was most important with him was how he dealt with World War II, because we were young children, I mean, the three of us, in 1939. So he ordered an air raid shelter and it couldn't fit into the house. So he just went into the garden and dug under a flowerbed and put an air raid shelter there. So then he built a little playhouse for us in the garden, and he built a telephone, his own version of a telephone, and he had a telephone to the air raid shelter and a telephone to the house. So we would go and be put to sleep in the air raid shelter under the flowerbeds and we would call him when we could hear the German planes because they sounded different.

[00:05:09]
They handled the gasoline in a different way. So while an English plane would go *zzzzzz*, a German plane would go *zzir-zzir-zzir-zzir*, so you could tell whenever you were about to be bombed. We spent a lot of time during the war in that air raid shelter. But then they would leave about three in the morning because they had to get back over the English Channel before

[01:38:41]
BOYTIM: Well, could you elaborate more on your undergraduate research, your thesis?

[01:38:46]
GLUSKER: Sorry?

[01:38:46]
BOYTIM: Could you elaborate more on your undergraduate research for your thesis? I think you worked with Harold—

[01:38:52]
GLUSKER: Well, I was working out the structure of vitamin B_{12}.

[01:38:56]
BOYTIM: But first with Harold [Warris] Thompson, you were doing infrared spectroscopy.

[01:39:00]
GLUSKER: Oh, that. Yes, yes. I'm sorry. Oh, yes.

[01:39:04]
BOYTIM: Getting ahead of ourselves.

[01:39:06]
GLUSKER: I had heard him lecture. You do have to—yes, you have to do that. I heard him lecture. He was very clear about what he talked about, and I thought, "This is really good." He was quite a famous footballer who nobody liked. So I said I would like to work with him. So he came and he, sort of, looked me over and said, "Fine." Said, "I've never had a woman." So there were twelve men and me. That was my introduction to more than chemistry. But it worked out very well because the men were delighted to find there was someone who would sew on their buttons and then they would . . . . We became great friends, and one or two of them were married, but only one or two, and the rest would ask me to go do things with them. So I got to know them, and that's where I met my husband, because he was one of the people in that lab.

[01:40:20]
He [Harold Thompson] was not used to dealing with women. Of course, he was always trying to persuade me that I could go and do something with him, my boss, which was totally

inappropriate, of course. So keeping him at bay was quite a lesson, and not very pleasant. They thought I might do a mixture of spectroscopy and crystallography for my doctorate. But I said finally, "I just don't want any more to do with him." I just stayed with Dorothy Hodgkin. But I did get to publish, yes, I got to publish some papers on what . . . methyl . . . oh, yes, deuterium chloride and so on, and get the bond length. So you can measure the bond length to great accuracy. So you know a lot about deuterium and chlorine and then you could compare [END OF AUDIO, FILE 1.1] it with hydrogen, which is—deuterium and hydrogen are the same kind of element and, sort of, write a little bit about that. But then I went on to more complicated molecules, and you don't find out as much as you hope you will. So I thought, "Well, crystallography, you do find out a lot more." So I said I wanted to work with Dorothy and she was very pleased and she said she got a place for me because she wanted to be working on B_{12}.

[00:00:30]
So it was, kind of, you know, my luck came. And I'm telling you, my life has been luck. Luck came in again and I got this great problem, which was the largest structure done at that time. So everyone was very excited about it. So all these people would come and see me and I got to give the first talk on it. People at Cambridge wanted to say that they had made the crystals that I worked on, and therefore, they were the ones who should get the honor for the structure. They didn't know anything about it chemically. That was pretty hard work, but it was good. In order to find it, you have to use your brain and develop methods of doing things.

[00:01:25]
And then I got connected with the man [Kenneth N. Trueblood] at UCLA [University of California, Los Angeles] in California, a professor there, and he was working with the main, huge computer they had. So they could do the calculations, some huge calculations. So we corresponded for years and we eventually wrote a textbook together.

[00:01:48]
BOYTIM: Yeah, that's Kenneth Trueblood?

[00:01:51]
GLUSKER: Yeah, Trueblood, yes. So yes, I mean, working on vitamin B_{12} was great. As they say, it was the largest at the time, and that's what—one of the things Dorothy got a Nobel Prize for, for penicillin.

[00:02:17]
SCHNEIDER: So you talked a little bit about your lab and some of the men in the lab who you became friends with. What kinds of things would you all do? Would you socialize in the lab? Would you, after you did the lab work, go elsewhere and spend time together? What was that social life like?

[00:02:41]
GLUSKER: Well, I would go in in the morning when Don [Donald L. Glusker], who was eventually my husband, would be—he was working on some other project to do with charge-transfer molecules, I think. But he had to set up the apparatus. And it took—in those days, the apparatus took an hour to get itself together in the morning. After you turn it on, you don't . . . . So he would sit and chat with me and tell me what life was like in America and what kind of a girl he was going to marry and what he would get for her. Never asked me out. But we would chat every morning for an hour, so we got to know each other very well.

[00:03:28]
And then the others, well, we would go to have lunch together sometimes, but not always. They would invite me out to dinner once in a while, you know, in a restaurant. Nothing very fancy, just a student restaurant, but they would pick up the bill, usually give me flowers or something. There was a couple who were married who didn't do that, of course, but we were friends. I was friends of them and their wives. Sometimes we could go on the river in the afternoon together. But it all worked out. I mean, I was friends with all of them.

[00:04:18]
And a lot of them have come here. If they were ever in America, they would make a point to come and see me and tell me what their lives were like. They'd got married and had kids, here are pictures of their children, and what were my children like, and so on. It was a good relationship. I wasn't really dating. I was just interested in getting on with chemistry. So I wasn't really very—I did have some dates, yes, but not too many. They were not in the lab. I had one date for a while, it was fairly serious for a while, but then it was obvious it wasn't going to work out.

[00:05:02]
And you were not allowed to be married when you were in college. There was a girl in my year who was a great mathematician and she got married—no, she wanted to get married and she got pregnant anyway and then, in those days, you got married. So she had to get married. She had to choose between marriage or getting her degree. She finally chose marriage and I always felt that was really very unfair. She was in her last year, so what, but . . . . So yes. A lot of friends, a lot of very good men friends and their wives. As I say, they would come here. One [Ronald T. Reuther] was the head of the zoo here in Philadelphia [Philadelphia Zoo] for a while.

[00:05:52]
BOYTIM: Oh, really? Huh.

[00:05:52]
GLUSKER: Ron Reuther, yes. Married an English girl. I think he arranged for the two of us to

go for dinner with him in some part of the zoo. That was good. But then he had a fight with the board of the zoo and had to go off and do something else.

[00:06:17]
BOYTIM: Were you thinking about your future and whether you would have to make a choice between marriage and a family or pursuing your doctorate?

[00:06:26]
GLUSKER: I wasn't really thinking about getting married at that point. I was in my early twenties and hadn't really thought much about it. Most people—I mean, for example, my uncle, my father's older brother, would, sort of, say, "You know, you're not bad looking. It's a shame you won't get married." And so I said to my mother, "Why won't I get married?" And she says, "Oh, because you're a scientist. They don't get married." So there we are. But no, as I say, I had—various men would ask me out and we would have good times together. I mean, go into the country or go—walks in the woods, that sort of thing.

[00:07:16]
I wasn't really thinking about it until just at the very end, when I was going around with Don after—I was doing graduate work in a different lab from him. He was a Rhodes Scholar, and so he had two years and he got an extra year. He loved Oxford and he loved going to Europe. And then we decided we would go and we thought we'd better get married, and my parents said, "Marrying an American is a big deal." Well, it was in those days. They adjusted in the end. No, I hadn't thought about getting married. I thought I wanted to make sure I got a good job after I finished. That was the main thought. Yes.

[00:08:14]
BOYTIM: And at what—

[00:08:14]
GLUSKER: A good job where I could do research. And I got job offers from both DuPont [E.I. du Pont de Nemours & Company] and Rohm and Haas [Company]. But then I was told [A. Lindo] Patterson was here in the Fox Chase [Cancer Center]. So I went to see him. He didn't have any money, he said, but I could be a technician for a year. So I was a technician for a year. Then he said, "We'll give you a proper title after that. I'll make sure of it," which he did. So new directors have all said, "Here's someone who rose from being a technician to being head of a lab." And I said, "No, I wasn't really a technician. It was only because he didn't have any money and I wanted desperately to work for him." He gave me a little salary, not a great one, and it was perfectly adequate. I mean, in those days, two good salaries really made things work. Yes.

[00:09:18]
BOYTIM: If you were imagining going on to a career in research, I assume you always figured you would go on to get a PhD after.

[00:09:28]
GLUSKER: Yes, yes, yes. Well, that's what I was doing. The B_{12} was in the PhD. That worked out well. So that's when I got to . . . well, you're going to talk about it another day, right?

[00:09:45]
BOYTIM: Yeah. I guess . . . we've been talking for a couple of hours. I don't know if there are any other questions from undergraduate years or anything else you want to cover, Sarah?

[00:09:55]
SCHNEIDER: I think this could be a good place to stop and then pick back up.

[00:09:59]
GLUSKER: If you think of another undergraduate question, you could sneak them in.

[00:10:02]
BOYTIM: Of course. We'll do that. I'll stop the recording.

[END OF AUDIO, FILE 1.2]

[END OF INTERVIEW]

INTERVIEWEE: **Jenny Pickworth Glusker**

INTERVIEWERS: **Jacqueline Boytim**
Sarah Schneider

LOCATION: **Interviewee's home**
Huntingdon Valley, Pennsylvania

DATE: **3 November 2022**

[00:00:05]
SCHNEIDER: So today is Thursday, November 3, 2022. My name is Sarah Schneider and I am joined by Jacqueline Boytim, and we're here with Dr. Jenny Glusker [at her home in Huntingdon Valley, Pennsylvania] to conduct the second session of the oral history interview.

[00:00:18]
BOYTIM: Thanks, Sarah. So, Jenny, yesterday we were talking about your undergraduate research at Oxford, which you did under Harold Thompson getting into IR spectroscopy. You talked a little bit about the environment in that lab. It was an all-male lab, and they weren't familiar with—

[00:00:37]
GLUSKER: It was an all-male lab. And everyone was watching to see how the men would react to having a woman. But they—I could get along fairly well with all of them and managed to flood the lab once, but that's about all I did that was bad.

[00:00:56]
BOYTIM: Oh, no. How did that happen?

[00:00:59]
GLUSKER: Oh, it's very easy when you're doing chemistry. You put your pipe into the glasswork and it springs out on the water pressure. So they all helped. Most of them still, during the time since, have been in communication, you know, tell me who they married and how many children they had and what their children were doing. So we exchanged notes. So I've kept contact with, I'd say, most of them. Some of them have done very well, and some of them teach other parts of science. Most of them are in England or Scotland. But it's nice to hear [from] them. I got asked to give a talk in Edinburgh once and I went up, and my husband went

with me, too, and there, as one of the professors, was somebody who had been in the lab. So we could catch up, all of a sudden, unexpectedly.

[00:02:03]
BOYTIM: That's nice.

[00:02:04]
GLUSKER: So that was nice. And that was the place that had a letter from [Dmitri Ivanovich] Mendeleev who designed the periodic table. So I said, "Please, please, let me touch the letter." [laughter] Ridiculous, but I've touched a letter that Mendeleev touched.

[00:02:26]
BOYTIM: Part of history.

[00:02:28]
GLUSKER: So, yes, I always liked chemistry. Yes.

[00:02:32]
BOYTIM: I guess we talked a little bit about this yesterday, I wasn't sure how much to go back into, but your research was on IR spectroscopy. You were looking at deuterium chloride with that.

[00:02:42]
GLUSKER: Yes.

[00:02:43]
BOYTIM: Do you want to talk through maybe the state of the field at the time? My understanding is that IR spectroscopy, kind of, took off in the 1940s with advanced instrumentation.

[00:02:55]
GLUSKER: Yes. Well, if you have something like deuterium chloride, you can tell what its bond length is and variations with respect to temperature or something and compare it with where the deuterium is replaced by hydrogen or the other way around. And then when you get to a more complicated molecule, you don't really have measurements on everything. You

measure the moments of inertia, and from that you can derive something about the distances. But I thought it was rather unsatisfactory.

[00:03:32]
The problem with the professors is, most of them just thought women were a nuisance and you didn't need them and they didn't do very well anyway. You didn't get too much support except for one or two who thought it was interesting and would help along, and I felt I could go to. The person I was working for was very good in his science, but he was not a person you would tell if you were having problems, you know, or anything. You would go to one of these other professors and they'd say, "How's it going? Can I help you at all?"

[00:04:20]
I did a year's research in spectroscopy and went to a meeting in Paris and met some of the early workers in that field. That was interesting. You get to know who people are and they get to know who I am, what I've done. I could, sort of, see a different way of doing an analysis, so that was a help. Went on to the methyl halides and found that they were more difficult to analyze. So I finally left. But I got a thesis that went in. So I got a good degree. Yes.

[00:05:14]
BOYTIM: So at what point did you start talking with Dorothy Hodgkin about working with her for your graduate . . . ?

[00:05:19]
GLUSKER: Well, she was my tutor, you see, in college. And so I saw her regularly, and she was very interested in what I was doing in spectroscopy. I mean, just that I was doing it and that I was doing it properly and being careful about—you're dealing with lots and lots of numbers. I was trained from the beginning, you always have to go back and check your numbers. It might take a day to do so, but you have to do it anyway. And you'll be asked, "Have you checked your numbers or did you just write it down?"

[00:05:58]
BOYTIM: Was she one of the . . . ?

[00:06:00]
GLUSKER: So she knew I was a fairly careful worker and she said what I . . . . The idea was it might be interesting to look at some crystal structures where you have big planar molecules and then look at the spectroscopy and see if the analysis methods at the time gave the same answer for how the flat molecule lay. And she thought that might be a nice subject for a doctorate. So I thought, yes, but then I was having trouble with my boss who was a little bit too affectionate and so on. I had to keep him at a distance. So I decided I just couldn't deal with that. So I said,